

Brett Loy

Foster City | 650-766-9194 | brettloy07@gmail.com | LinkedIn: Brett Loy

EDUCATION

University of Southern California

Master of Science in Applied Analytics

Present – 05/2028

Foster City, CA/ Remote

- **Coursework:** Applied Business Analytics and AI

University of California, Davis

Bachelor of Science in Statistics – Data Science Track, Minor in Mathematics

09/2021 – 06/2025

Davis, CA

- **Coursework:** Practice in Data Science, Multivariate Data Analysis, Statistical Learning, Applied Linear Algebra

SKILLS

- Software Languages: Python(Pandas, NumPy, Scikit-Learn), R(dplyr, tidyr, ggplot2), SQL
- Technical Skills: Tableau, Microsoft Office, Google Workspace, Machine Learning Models, Data Pipelines

SELECTED DATA SCIENCE PROJECTS

Airbnb Market Trends Analysis | *Python(pandas), SQL, Tableau*

05/2025 – 07/2025

- Performed data wrangling processes on Airbnb listings CSV dataset using SQL, including column selection, null value imputation, data type conversions, and outlier detection or removal to ensure data quality.
- Engineered SQL queries to derive actionable insights on pricing trends, neighborhood segmentation, and room type distributions, aggregating metrics for EDA and trend identification.
- Built an interactive Tableau dashboard featuring dynamic visualizations of average pricing, review scores, and property availability filtered by neighborhood and room type, facilitating market trend exploration.
- Executed end-to-end data analysis pipeline including data cleaning, querying, and interactive data visuals.

MLB Pitcher Analysis | *R(dplyr, ggplot2)*

12/2024 – 02/2025

- Leveraged baseballr package to wrangle and clean variables including pitch type, release speed, hit events, etc; conducted EDA via descriptive data analytics to compute averages and frequencies to display robust analysis.
- Conducted K-means clustering on multi-dimensional pitch data to identify 3 optimal clusters, but revealing Cluster 2 (high-speed fastballs with vertical movement) as most effective with lowest 19.4% hit rate.
- Derived insights from data patterns and created visualizations such as boxplots, bar charts, heatmaps, scatterplots, and hit rate/cluster trend analyses to illustrate pitch evolution and strategic effectiveness over time.

Restaurant Revenue Prediction | *Python(pandas, numpy, matplotlib)*

06/2024 – 08/2024

- Built machine learning predictive models using regression and classification methods such as linear and logistic regression, LDA, QDA, and SVM to forecast restaurant revenue and achieved up to 94.6% accuracy.
- Applied resampling methods including cross-validation and bootstrapping to evaluate the model's performance based on stability, variability, and ability to generalize to unseen datasets for enhanced predictive ability.
- Developed interpretable data visualizations of classification decision boundaries to uncover insights, analyzing how key business metrics and predictive models differentiate high and low performing restaurants.

OTHER ANALYTICS AND LEADERSHIP EXPERIENCE

Entrepreneur

Prime Cuts Barber Company

03/2020 – Present

Foster City/Davis, CA

- Integrated an online booking system that automated scheduling and generated structured datasets for analysis.
- Conducted EDA on demand patterns using Excel(pivot tables) and Python for data manipulation, uncovering peak hours and client behavior trends to improve marketing strategy leading 15% increase in monthly revenue.

VP of Finance

Tau Kappa Epsilon Fraternity

09/2022 – 06/2025

Davis, CA

- Built Excel financial trackers using VLOOKUP and pivot tables to simplify event reimbursements and expense logging, helping executive team quickly read the budget without needing deep financial expertise.
- Managed an annual budget of \$10,000 for philanthropy and over \$45,000 in house rent, overseeing all financial transactions, expense tracking, and budget allocation to ensure responsible chapter spending.